Supplementary Material of UCA-GAN

A Explanation of $p'(\boldsymbol{x}, l)$

We give an explanation of $p'(\boldsymbol{x}, l)$. First, for distribution $p_{\text{real}}(\boldsymbol{x}, l)$, there are two cases: normal and abnormal. In the case of abnormal samples, according to the Eq. (4), there are two cases: samples drawn from the auxiliary data source distribution p^a_{real} and samples drawn from the distribution p^b_{real}. Therefore, the samples following the distribution $p_{\text{real}}(\boldsymbol{x}, l)$ fall into three categories: normal, anomalous and originating from p^a_{real}, as well as anomalous and originating from p^b_{real}, with probabilities of $1-\gamma_p$, $\gamma_p\gamma_{\text{aux}}$, and $\gamma_p(1-\gamma_{\text{aux}})$, respectively. Taking only the first and third cases, the sample's $\boldsymbol{x}$ and label l actually follow the distribution $p'(\boldsymbol{x}, l)$.

B Proof of Theorem 1

This section proves Theorem 1. Part of this proof draws on the work of Su et al. [25]. Because the theorem in that work are based on its model, we do not directly use the theorem here, but rather organize the reasoning. The proof is below.

Proof. First, assume that Condition 1, Condition 2 and Condition 3 are true, and then do the subsequent reasoning. Define $p_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) = p_{\text{real}}(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}; \boldsymbol{\theta}_{\text{enc}})$, and define $p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) = p^a_{\text{real}}(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x})$. And then there is

$$\begin{aligned}
p_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) = \ & p_{\text{real}}(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) \\
= \ & (1-\gamma_p)p_{\text{real}}(\boldsymbol{x}|l=0)p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) + \gamma_p p_{\text{real}}(\boldsymbol{x}|l=1)p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) \\
= \ & (1-\gamma_p)p_{\text{real}}(\boldsymbol{x}|l=0)p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) + \gamma_p\gamma_{\text{aux}}p^a_{\text{real}}(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) \\
& + \gamma_p(1-\gamma_{\text{aux}})p^b_{\text{real}}(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) \\
= \ & \gamma_p\gamma_{\text{aux}}p^a_{\text{real}}(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) + (1-\gamma_p)p_{\text{real}}(\boldsymbol{x}|l=0)p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) \\
& + \gamma_p(1-\gamma_{\text{aux}})p^b_{\text{real}}(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) \\
= \ & \gamma_p\gamma_{\text{aux}}p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}})(p'(l=0)p'(\boldsymbol{x}|l=0)p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) \\
& + p'(l=1)p'(\boldsymbol{x}|l=1)p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x})) \\
= \ & \gamma_p\gamma_{\text{aux}}p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}})p'(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x})
\end{aligned} \tag{15}$$

Define $p'(\boldsymbol{x}, \boldsymbol{z}) = p'(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x})$, then the above expression becomes

$$p_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) = \gamma_p\gamma_{\text{aux}}p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}})p'(\boldsymbol{x}, \boldsymbol{z}) \tag{16}$$

Later, $p_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})$ is abbreviated as p_{enc}, $p_{\text{gen}}(\boldsymbol{x}, \boldsymbol{z})$ is abbreviated as p_{gen}, $p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})$ is abbreviated as p^a_{enc}, $D(\boldsymbol{x}, \boldsymbol{z})$ is abbreviated as D. In the Eq. (5),

$$V(D) = \iint_S ((D-1)^2 p_{\text{enc}} + (D-0)^2 p_{\text{gen}} + (D-d)^2 p^a_{\text{enc}}) \mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} \tag{17}$$

Here, S is the universal set of values for $(\boldsymbol{x},\boldsymbol{z})$. For any $(\boldsymbol{x},\boldsymbol{z}) \in S$, define the function $f(u) = (u-1)^2 p_{\text{enc}} + (u-0)^2 p_{\text{gen}} + (u-d)^2 p^a_{\text{enc}}$. Then the function f is a quadratic function that opens upwards, and thus takes its minimum value at the zero of its derivative f'. The derivative of f is

$$f'(u) = (2p_{\text{enc}} + 2p_{\text{gen}} + 2p^a_{\text{enc}})u - 2p_{\text{enc}} - 2dp^a_{\text{enc}} \tag{18}$$

Thus, the zero of f' is $\frac{p_{\text{enc}}+dp^a_{\text{enc}}}{p_{\text{enc}}+p_{\text{gen}}+p^a_{\text{enc}}}$. If we set $D(\boldsymbol{x},\boldsymbol{z}) = \frac{p_{\text{enc}}+dp^a_{\text{enc}}}{p_{\text{enc}}+p_{\text{gen}}+p^a_{\text{enc}}}$, then $V(D)$ attains its minimum, and it is easy to show that this is the unique minimizer. Substituting Eq. (16) into $D(\boldsymbol{x},\boldsymbol{z}) = \frac{p_{\text{enc}}+dp^a_{\text{enc}}}{p_{\text{enc}}+p_{\text{gen}}+p^a_{\text{enc}}}$ yields

$$D(\boldsymbol{x},\boldsymbol{z}) = \frac{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}}}{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}}} \tag{19}$$

Substituting the Eq. (16) and Eq. (19) into the expression of V in Eq. (6), we get

$$\begin{aligned} V(\boldsymbol{\theta}_{\text{pz}},\boldsymbol{\theta}_{\text{gen}},\boldsymbol{\theta}_{\text{enc}}) = \iint_S &\left(\frac{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}}}{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}}} - c\right)^2 \\ &\cdot (\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} \end{aligned} \tag{20}$$

Define the function $\varphi(u) = (u-c)^2$, then

$$\begin{aligned} V(\boldsymbol{\theta}_{\text{pz}},\boldsymbol{\theta}_{\text{gen}},\boldsymbol{\theta}_{\text{enc}}) = 3\iint_S &\varphi\left(\frac{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}}}{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}}}\right) \\ &\cdot \frac{1}{3}(\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} \end{aligned} \tag{21}$$

Since φ is a convex function, by Jensen's inequality $\int_\Omega \varphi(g(\boldsymbol{v}))p(\boldsymbol{v})\mathrm{d}\boldsymbol{v} \geq \varphi(\int_\Omega g(\boldsymbol{v})p(\boldsymbol{v})\mathrm{d}\boldsymbol{v})$, we have

$$\begin{aligned} V(\boldsymbol{\theta}_{\text{pz}},\boldsymbol{\theta}_{\text{gen}},\boldsymbol{\theta}_{\text{enc}}) \geq\ & 3\varphi\Big(\iint_S \frac{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}}}{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}}} \\ &\cdot \frac{1}{3}(\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z}\Big) \\ =\ & 3\varphi\Big(\frac{1}{3}\iint_S (\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z}\Big) \\ =\ & 3\varphi\Big(\frac{1}{3}(\gamma_p\gamma_{\text{aux}}\iint_S p^a_{\text{enc}}\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} + (1-\gamma_p\gamma_{\text{aux}})\iint_S p'\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} \\ &+ d\iint_S p^a_{\text{enc}}\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z})\Big) \\ =\ & 3\varphi\Big(\frac{1}{3}(1+d)\Big) \end{aligned} \tag{22}$$

The necessary and sufficient condition for the equality of $\geq$ to hold is that the following function with respect to $(\boldsymbol{x},\boldsymbol{z})$

$$\frac{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}}(\boldsymbol{x},\boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}})p'(\boldsymbol{x},\boldsymbol{z}) + dp^a_{\text{enc}}(\boldsymbol{x},\boldsymbol{z})}{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}}(\boldsymbol{x},\boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}})p'(\boldsymbol{x},\boldsymbol{z}) + p_{\text{gen}}(\boldsymbol{x},\boldsymbol{z}) + p^a_{\text{enc}}(\boldsymbol{x},\boldsymbol{z})} \tag{23}$$

is a constant function. Because $d = \frac{1-2\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}}$,

$$3\varphi(\frac{1}{3}(1+d)) = 3\varphi(\frac{1-\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}}) \tag{24}$$

Thus,

$$V(\boldsymbol{\theta}_{\text{pz}}, \boldsymbol{\theta}_{\text{gen}}, \boldsymbol{\theta}_{\text{enc}}) \geq 3\varphi(\frac{1-\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}}) \tag{25}$$

The right-hand side of the inequality is independent of the learning parameters, hence it is a lower bound of the function to be optimized. Moreover, the necessary and sufficient condition for the inequality to become an equality is as given earlier. According to Condition 1, we assume that when $\boldsymbol{\theta}_{\text{pz}}$, $\boldsymbol{\theta}_{\text{gen}}$, and $\boldsymbol{\theta}_{\text{enc}}$ take the values $\hat{\boldsymbol{\theta}}_{\text{pz}}$, $\hat{\boldsymbol{\theta}}_{\text{gen}}$, and $\hat{\boldsymbol{\theta}}_{\text{enc}}$ respectively, we have $p_{\text{gen}}(\boldsymbol{x}, l) = p'(\boldsymbol{x}, l)$ and $p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}) = p_{\text{gen}}(\boldsymbol{z}|\boldsymbol{x})$, that is,

$$p_{\text{gen}}(\boldsymbol{x}, l; \hat{\boldsymbol{\theta}}_{\text{pz}}, \hat{\boldsymbol{\theta}}_{\text{gen}}) = p'(\boldsymbol{x}, l) \tag{26}$$

$$p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}; \hat{\boldsymbol{\theta}}_{\text{enc}}) = p_{\text{gen}}(\boldsymbol{z}|\boldsymbol{x}; \hat{\boldsymbol{\theta}}_{\text{pz}}, \hat{\boldsymbol{\theta}}_{\text{gen}}) \tag{27}$$

Then $p_{\text{gen}}(\boldsymbol{x}; \hat{\boldsymbol{\theta}}_{\text{pz}}, \hat{\boldsymbol{\theta}}_{\text{gen}}) = \sum_{l=0}^{1} p_{\text{gen}}(\boldsymbol{x}, l; \hat{\boldsymbol{\theta}}_{\text{pz}}, \hat{\boldsymbol{\theta}}_{\text{gen}}) = \sum_{l=0}^{1} p'(\boldsymbol{x}, l) = p'(\boldsymbol{x})$, so

$$\begin{aligned} p_{\text{gen}}(\boldsymbol{x}, \boldsymbol{z}; \hat{\boldsymbol{\theta}}_{\text{pz}}, \hat{\boldsymbol{\theta}}_{\text{gen}}) &= p_{\text{gen}}(\boldsymbol{x}; \hat{\boldsymbol{\theta}}_{\text{pz}}, \hat{\boldsymbol{\theta}}_{\text{gen}}) p_{\text{gen}}(\boldsymbol{z}|\boldsymbol{x}; \hat{\boldsymbol{\theta}}_{\text{pz}}, \hat{\boldsymbol{\theta}}_{\text{gen}}) \\ &= p'(\boldsymbol{x}) p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x}; \hat{\boldsymbol{\theta}}_{\text{enc}}) \\ &= p'(\boldsymbol{x}, \boldsymbol{z}; \hat{\boldsymbol{\theta}}_{\text{enc}}) \end{aligned} \tag{28}$$

So, when $\boldsymbol{\theta}_{\text{pz}}$, $\boldsymbol{\theta}_{\text{gen}}$, and $\boldsymbol{\theta}_{\text{enc}}$ take the values $\hat{\boldsymbol{\theta}}_{\text{pz}}$, $\hat{\boldsymbol{\theta}}_{\text{gen}}$, and $\hat{\boldsymbol{\theta}}_{\text{enc}}$, respectively,

$$\begin{aligned} &\frac{\gamma_p\gamma_{\text{aux}} p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}}) p'(\boldsymbol{x}, \boldsymbol{z}) + d p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})}{\gamma_p\gamma_{\text{aux}} p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}}) p'(\boldsymbol{x}, \boldsymbol{z}) + p_{\text{gen}}(\boldsymbol{x}, \boldsymbol{z}) + p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})} \\ &= \frac{\gamma_p\gamma_{\text{aux}} p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}}) p'(\boldsymbol{x}, \boldsymbol{z}) + \frac{1-2\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}} p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})}{\gamma_p\gamma_{\text{aux}} p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z}) + (1-\gamma_p\gamma_{\text{aux}}) p'(\boldsymbol{x}, \boldsymbol{z}) + p'(\boldsymbol{x}, \boldsymbol{z}) + p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})} \\ &= \frac{1-\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}} \end{aligned} \tag{29}$$

is a constant function, so at this point, equality holds in Eq. (25), and thus, $V(\boldsymbol{\theta}_{\text{pz}}, \boldsymbol{\theta}_{\text{gen}}, \boldsymbol{\theta}_{\text{enc}})$ attains a minimum. This proves the existence of an optimal solution in Theorem 1, and when the minimum is attained, equality holds in (25). Suppose that $V(\boldsymbol{\theta}_{\text{pz}}, \boldsymbol{\theta}_{\text{gen}}, \boldsymbol{\theta}_{\text{enc}})$ attains its minimum when $\boldsymbol{\theta}_{\text{pz}}$, $\boldsymbol{\theta}_{\text{gen}}$, and $\boldsymbol{\theta}_{\text{enc}}$ take the values $\boldsymbol{\theta}^*_{\text{pz}}$, $\boldsymbol{\theta}^*_{\text{gen}}$, and $\boldsymbol{\theta}^*_{\text{enc}}$, respectively, where $\boldsymbol{\theta}^*_{\text{pz}}$, $\boldsymbol{\theta}^*_{\text{gen}}$, and $\boldsymbol{\theta}^*_{\text{enc}}$ are arbitrary. Then, when $\boldsymbol{\theta}_{\text{pz}}$, $\boldsymbol{\theta}_{\text{gen}}$, and $\boldsymbol{\theta}_{\text{enc}}$ take the values $\boldsymbol{\theta}^*_{\text{pz}}$, $\boldsymbol{\theta}^*_{\text{gen}}$, and $\boldsymbol{\theta}^*_{\text{enc}}$, respectively, equality holds in Eq. (25). According to the equality's necessary and sufficient condition, we obtain

$$\frac{\gamma_p\gamma_{\text{aux}} p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}}) p' + d p^a_{\text{enc}}}{\gamma_p\gamma_{\text{aux}} p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}}) p' + p_{\text{gen}} + p^a_{\text{enc}}} = b \tag{30}$$

where b is a constant that is independent of $\boldsymbol{x}$ and $\boldsymbol{z}$. So we have

$$\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}} = b\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + b(1-\gamma_p\gamma_{\text{aux}})p' + bp_{\text{gen}} + bp^a_{\text{enc}} \tag{31}$$

Integrating both sides, we obtain

$$\begin{aligned}&\iint_S(\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z}\\ &= \iint_S(b\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + b(1-\gamma_p\gamma_{\text{aux}})p' + bp_{\text{gen}} + bp^a_{\text{enc}})\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z}\end{aligned} \tag{32}$$

Then,

$$\begin{aligned}&\gamma_p\gamma_{\text{aux}}\iint_S p^a_{\text{enc}}\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} + (1-\gamma_p\gamma_{\text{aux}})\iint_S p'\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} + d\iint_S p^a_{\text{enc}}\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z}\\ &= b\gamma_p\gamma_{\text{aux}}\iint_S p^a_{\text{enc}}\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} + b(1-\gamma_p\gamma_{\text{aux}})\iint_S p'\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} + b\iint_S p_{\text{gen}}\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z} + b\iint_S p^a_{\text{enc}}\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{z}\end{aligned} \tag{33}$$

Then,

$$\gamma_p\gamma_{\text{aux}} + (1-\gamma_p\gamma_{\text{aux}}) + d = b\gamma_p\gamma_{\text{aux}} + b(1-\gamma_p\gamma_{\text{aux}}) + b + b \tag{34}$$

Then,

$$b = \frac{1}{3}(1+d) = \frac{1-\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}} \tag{35}$$

Thus,

$$\frac{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + dp^a_{\text{enc}}}{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}}} = \frac{1-\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}} \tag{36}$$

Substituting $d = \frac{1-2\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}}$ into this equation, we get

$$\frac{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + \frac{1-2\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}}p^a_{\text{enc}}}{\gamma_p\gamma_{\text{aux}}p^a_{\text{enc}} + (1-\gamma_p\gamma_{\text{aux}})p' + p_{\text{gen}} + p^a_{\text{enc}}} = \frac{1-\gamma_p\gamma_{\text{aux}}}{2-\gamma_p\gamma_{\text{aux}}} \tag{37}$$

From this equation, we can obtain

$$p_{\text{gen}} = p' \tag{38}$$

that is,

$$p_{\text{gen}}(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p'(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}^*_{\text{enc}}) = p'(\boldsymbol{x})p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x};\boldsymbol{\theta}^*_{\text{enc}}) \tag{39}$$

Therefore, their marginal distributions and conditional distributions are the same, as follows:

$$p_{\text{gen}}(\boldsymbol{x};\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p'(\boldsymbol{x}) \tag{40}$$

$$p_{\text{gen}}(\boldsymbol{z}|\boldsymbol{x};\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x};\boldsymbol{\theta}^*_{\text{enc}}) \tag{41}$$

Because of

$$
\begin{aligned}
p_{\text{gen}}(\boldsymbol{x};\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p'(\boldsymbol{x}) = \sum_{l=0}^{1} p'(\boldsymbol{x},l) = \sum_{l=0}^{1} p_{\text{gen}}(\boldsymbol{x},l;\hat{\boldsymbol{\theta}}_{\text{pz}},\hat{\boldsymbol{\theta}}_{\text{gen}}) \\
= p_{\text{gen}}(\boldsymbol{x};\hat{\boldsymbol{\theta}}_{\text{pz}},\hat{\boldsymbol{\theta}}_{\text{gen}})
\end{aligned}
\tag{42}
$$

and Condition 3, we get

$$
p_{\text{gen}}(\boldsymbol{z};\boldsymbol{\theta}^*_{\text{pz}}) = p_{\text{gen}}(\boldsymbol{z};\hat{\boldsymbol{\theta}}_{\text{pz}}) \tag{43}
$$

$$
G(\boldsymbol{z};\boldsymbol{\theta}^*_{\text{gen}}) = G(\boldsymbol{z};\hat{\boldsymbol{\theta}}_{\text{gen}}) \tag{44}
$$

According to Eq. (1) and Eq. (2) in the model setup, the marginal distribution $p_{\text{gen}}(\boldsymbol{z};p_a,\boldsymbol{\mu}_a)$ of the model is obtained as

$$
\begin{aligned}
p_{\text{gen}}(\boldsymbol{z};p_a,\boldsymbol{\mu}_a) &= \sum_{l=0}^{1} p(\boldsymbol{z},l) \\
&= \sum_{l=0}^{1} P(l)p(\boldsymbol{z}|l) \\
&= (1-p_a)N(\boldsymbol{z}|\mathbf{0},\mathbf{I}) + p_a N(\boldsymbol{z}|\boldsymbol{\mu}_a,\sigma_a^2\mathbf{I})
\end{aligned}
\tag{45}
$$

Let $\boldsymbol{\theta}^*_{\text{pz}} = (p^*_a,\boldsymbol{\mu}^*_a)$, $\hat{\boldsymbol{\theta}}_{\text{pz}} = (\hat{p}_a,\hat{\boldsymbol{\mu}}_a)$, then Eq. (43) is that

$$
(1-p^*_a)N(\boldsymbol{z}|\mathbf{0},\mathbf{I}) + p^*_a N(\boldsymbol{z}|\boldsymbol{\mu}^*_a,\sigma_a^2\mathbf{I}) = (1-\hat{p}_a)N(\boldsymbol{z}|\mathbf{0},\mathbf{I}) + \hat{p}_a N(\boldsymbol{z}|\hat{\boldsymbol{\mu}}_a,\sigma_a^2\mathbf{I}) \tag{46}
$$

From Eq. (46), it is easy to obtain $p^*_a = \hat{p}_a$, $\boldsymbol{\mu}^*_a = \hat{\boldsymbol{\mu}}_a$, and thus $\boldsymbol{\theta}^*_{\text{pz}} = \hat{\boldsymbol{\theta}}_{\text{pz}}$. Therefore, $p_{\text{gen}}(\boldsymbol{z},l;\boldsymbol{\theta}^*_{\text{pz}}) = p_{\text{gen}}(\boldsymbol{z},l;\hat{\boldsymbol{\theta}}_{\text{pz}})$. Also, because of Eq. (44), i.e., $p_{\text{gen}}(\boldsymbol{x}|\boldsymbol{z};\boldsymbol{\theta}^*_{\text{gen}}) = p_{\text{gen}}(\boldsymbol{x}|\boldsymbol{z};\hat{\boldsymbol{\theta}}_{\text{gen}})$,

$$
\begin{aligned}
p_{\text{gen}}(\boldsymbol{x},\boldsymbol{z},l;\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) &= p_{\text{gen}}(\boldsymbol{z},l;\boldsymbol{\theta}^*_{\text{pz}})p_{\text{gen}}(\boldsymbol{x}|\boldsymbol{z};\boldsymbol{\theta}^*_{\text{gen}}) \\
&= p_{\text{gen}}(\boldsymbol{z},l;\hat{\boldsymbol{\theta}}_{\text{pz}})p_{\text{gen}}(\boldsymbol{x}|\boldsymbol{z};\hat{\boldsymbol{\theta}}_{\text{gen}}) \\
&= p_{\text{gen}}(\boldsymbol{x},\boldsymbol{z},l;\hat{\boldsymbol{\theta}}_{\text{pz}},\hat{\boldsymbol{\theta}}_{\text{gen}})
\end{aligned}
\tag{47}
$$

Since these two joint distributions are the same, $p_{\text{gen}}(\boldsymbol{x},l;\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p_{\text{gen}}(\boldsymbol{x},l;\hat{\boldsymbol{\theta}}_{\text{pz}},\hat{\boldsymbol{\theta}}_{\text{gen}})$. Also, because of Eq. (26), $p_{\text{gen}}(\boldsymbol{x},l;\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p'(\boldsymbol{x},l)$, and thus, $p_{\text{gen}}(\boldsymbol{x}|l = 0;\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p'(\boldsymbol{x}|l=0)$. Furthermore, because of Eq. (9), $p_{\text{gen}}(\boldsymbol{x}|l=0;\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p'(\boldsymbol{x}|l=0) = p_{\text{real}}(\boldsymbol{x}|l=0)$. This proves $p_{\text{gen}}(\boldsymbol{x}|l=0;\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}}) = p_{\text{real}}(\boldsymbol{x}|l=0)$ in Theorem 1. Additionally, the preceding equation, Eq. (41), is $p_{\text{enc}}(\boldsymbol{z}|\boldsymbol{x};\boldsymbol{\theta}^*_{\text{enc}}) = p_{\text{gen}}(\boldsymbol{z}|\boldsymbol{x};\boldsymbol{\theta}^*_{\text{pz}},\boldsymbol{\theta}^*_{\text{gen}})$, which proves that the encoder distribution is the same as the posterior distribution of the probability model. Q.E.D.

C Detailed derivations of the training method's formulas

Regarding the learning steps of the discriminator, according to Eq. (5), we have

$$
\begin{aligned}
V(\boldsymbol{\theta}_d) = \;& \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-1)^2] \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\mathrm{gen}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-0)^2] \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p^a_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-d)^2] \\
= \;& \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-1)^2] \\
& + \int_{(\boldsymbol{x},\boldsymbol{z})} p_{\mathrm{gen}}(\boldsymbol{x},\boldsymbol{z})(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-0)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p^a_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-d)^2] \\
= \;& \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-1)^2] \\
& + \int_{(\boldsymbol{x},\boldsymbol{z})} (p(l=0)p(\boldsymbol{x},\boldsymbol{z}|l=0)+p(l=1)p(\boldsymbol{x},\boldsymbol{z}|l=1))(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-0)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p^a_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-d)^2] \\
= \;& \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-1)^2] \\
& + p(l=0)\int_{(\boldsymbol{x},\boldsymbol{z})} p(\boldsymbol{x},\boldsymbol{z}|l=0)(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-0)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + p(l=1)\int_{(\boldsymbol{x},\boldsymbol{z})} p(\boldsymbol{x},\boldsymbol{z}|l=1)(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-0)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p^a_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-d)^2] \\
= \;& \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-1)^2] \\
& + (1-p_a)\mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p(\boldsymbol{x},\boldsymbol{z}|l=0)}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-0)^2] \\
& + p_a\mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p(\boldsymbol{x},\boldsymbol{z}|l=1)}[(D(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_d)-0)^2] \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p^a_{\mathrm{enc}}(\boldsymbol{x},\boldsymbol{z})}[(D(\boldsymbol{x},\boldsymbol{z})-d;\boldsymbol{\theta}_d)^2] \\
\approx \;& \frac{1}{N_1}\sum_{i=1}^{N_1}(D(\tilde{\boldsymbol{x}}_i,\tilde{\boldsymbol{z}}_i;\boldsymbol{\theta}_d)-1)^2 \\
& + (1-p_a)\frac{1}{N_2}\sum_{i=1}^{N_2}(D(\tilde{\boldsymbol{x}}'_i,\tilde{\boldsymbol{z}}'_i;\boldsymbol{\theta}_d)-0)^2 \\
& + p_a\frac{1}{N_3}\sum_{i=1}^{N_3}(D(\tilde{\boldsymbol{x}}''_i,\tilde{\boldsymbol{z}}''_i;\boldsymbol{\theta}_d)-0)^2 \\
& + \frac{1}{N_4}\sum_{i=1}^{N_4}(D(\tilde{\boldsymbol{x}}'''_i,\tilde{\boldsymbol{z}}'''_i;\boldsymbol{\theta}_d)-d)^2
\end{aligned}
\tag{48}
$$

The parameter gradient is estimated based on this estimation formula, and then updated. $(\tilde{\boldsymbol{x}}_i,\tilde{\boldsymbol{z}}_i)$ are N_1 samples drawn randomly from the distribution

$p_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})$, where each sample is obtained by first randomly drawing a sample $\tilde{\boldsymbol{x}}_i$ from the contaminated dataset, and then setting $\tilde{\boldsymbol{z}}_i = E(\tilde{\boldsymbol{x}}_i)$. $(\tilde{\boldsymbol{x}}'_i, \tilde{\boldsymbol{z}}'_i)$ are N_2 samples drawn randomly from the distribution $p(\boldsymbol{x}, \boldsymbol{z}|l = 0)$, where each sample is obtained by first drawing $\tilde{\boldsymbol{z}}'_i$ from the Gaussian distribution $p(\boldsymbol{z}|l = 0) = N(\boldsymbol{z}|\mathbf{0}, \mathbf{I})$ using an algorithm, and then setting $\tilde{\boldsymbol{x}}'_i = G(\tilde{\boldsymbol{z}}'_i)$ to obtain $(\tilde{\boldsymbol{x}}'_i, \tilde{\boldsymbol{z}}'_i)$. $(\tilde{\boldsymbol{x}}''_i, \tilde{\boldsymbol{z}}''_i)$ are N_3 samples drawn randomly from the distribution $p(\boldsymbol{x}, \boldsymbol{z}|l = 1)$, where each sample is obtained by first drawing $\tilde{\boldsymbol{z}}''_i$ from the Gaussian distribution $p(\boldsymbol{z}|l = 1) = N(\boldsymbol{z}|\boldsymbol{\mu}_a, \sigma_a^2\mathbf{I})$ using an algorithm, and then setting $\tilde{\boldsymbol{x}}''_i = G(\tilde{\boldsymbol{z}}''_i)$ to obtain $(\tilde{\boldsymbol{x}}''_i, \tilde{\boldsymbol{z}}''_i)$. $(\tilde{\boldsymbol{x}}'''_i, \tilde{\boldsymbol{z}}'''_i)$ are N_4 samples drawn randomly from the distribution $p^a_{\text{enc}}(\boldsymbol{x}, \boldsymbol{z})$, where each sample is obtained by first randomly drawing a sample $\tilde{\boldsymbol{x}}'''_i$ from the auxiliary dataset, and then setting $\tilde{\boldsymbol{z}}'''_i = E(\tilde{\boldsymbol{x}}'''_i)$.

For the learning steps of the probabilistic model and encoder, according to Eq. (6), we have

$$\begin{aligned}
V(p_a, \boldsymbol{\mu}_a, \boldsymbol{\theta}_{\text{gen}}, \boldsymbol{\theta}_{\text{enc}}) = \; & \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\text{enc}}(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_{\text{enc}})}[(D(\boldsymbol{x},\boldsymbol{z})-c)^2] \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p_{\text{gen}}(\boldsymbol{x},\boldsymbol{z};p_a,\boldsymbol{\mu}_a,\boldsymbol{\theta}_{\text{gen}})}[(D(\boldsymbol{x},\boldsymbol{z})-c)^2] \\
& + \mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p^a_{\text{enc}}(\boldsymbol{x},\boldsymbol{z};\boldsymbol{\theta}_{\text{enc}})}[(D(\boldsymbol{x},\boldsymbol{z})-c)^2] \\
= \; & \mathbb{E}_{\boldsymbol{x}\sim p_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
& + \int_{(\boldsymbol{x},\boldsymbol{z})} p(\boldsymbol{x},\boldsymbol{z})(D(\boldsymbol{x},\boldsymbol{z})-c)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + \mathbb{E}_{\boldsymbol{x}\sim p^a_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
= \; & \mathbb{E}_{\boldsymbol{x}\sim p_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
& + \int_{(\boldsymbol{x},\boldsymbol{z})} (p(l=0)p(\boldsymbol{x},\boldsymbol{z}|l=0)+p(l=1)p(\boldsymbol{x},\boldsymbol{z}|l=1))(D(\boldsymbol{x},\boldsymbol{z})-c)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + \mathbb{E}_{\boldsymbol{x}\sim p^a_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
= \; & \mathbb{E}_{\boldsymbol{x}\sim p_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
& + p(l=0)\int_{(\boldsymbol{x},\boldsymbol{z})} p(\boldsymbol{x},\boldsymbol{z}|l=0)(D(\boldsymbol{x},\boldsymbol{z})-c)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + p(l=1)\int_{(\boldsymbol{x},\boldsymbol{z})} p(\boldsymbol{x},\boldsymbol{z}|l=1)(D(\boldsymbol{x},\boldsymbol{z})-c)^2 \mathrm{d}(\boldsymbol{x},\boldsymbol{z}) \\
& + \mathbb{E}_{\boldsymbol{x}\sim p^a_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
= \; & \mathbb{E}_{\boldsymbol{x}\sim p_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
& + (1-p_a)\mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p(\boldsymbol{x},\boldsymbol{z}|l=0)}[(D(\boldsymbol{x},\boldsymbol{z})-c)^2] \\
& + p_a\mathbb{E}_{(\boldsymbol{x},\boldsymbol{z})\sim p(\boldsymbol{x},\boldsymbol{z}|l=1)}[(D(\boldsymbol{x},\boldsymbol{z})-c)^2] \\
& + \mathbb{E}_{\boldsymbol{x}\sim p^a_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
= \; & \mathbb{E}_{\boldsymbol{x}\sim p_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
& + (1-p_a)\mathbb{E}_{\boldsymbol{z}\sim p(\boldsymbol{z}|l=0)}[(D(G(\boldsymbol{z}),\boldsymbol{z})-c)^2] \\
& + p_a\mathbb{E}_{\boldsymbol{z}\sim p(\boldsymbol{z}|l=1)}[(D(G(\boldsymbol{z}),\boldsymbol{z})-c)^2] \\
& + \mathbb{E}_{\boldsymbol{x}\sim p^a_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
= \; & \mathbb{E}_{\boldsymbol{x}\sim p_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2] \\
& + (1-p_a)\mathbb{E}_{\boldsymbol{z}\sim N(\boldsymbol{z}|\mathbf{0},\mathbf{I})}[(D(G(\boldsymbol{z}),\boldsymbol{z})-c)^2] \\
& + p_a\mathbb{E}_{\boldsymbol{z}\sim N(\boldsymbol{z}|\boldsymbol{\mu}_a,\sigma_a^2\mathbf{I})}[(D(G(\boldsymbol{z}),\boldsymbol{z})-c)^2] \\
& + \mathbb{E}_{\boldsymbol{x}\sim p^a_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x},E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}}))-c)^2]
\end{aligned} \tag{49}$$

Because the distribution of the third term of the final expression contains parameters to learn, the estimator that can estimate the gradients of the parameters can't be obtained directly. Thus, we reparameterize it [15]. If a random variable $\boldsymbol{t}$ follows the distribution $N(\boldsymbol{t}|\mathbf{0},\mathbf{I})$, $\sigma_a\boldsymbol{t}+\boldsymbol{\mu}_a$ follows the distribution $N(\cdot|\boldsymbol{\mu}_a,\sigma_a^2\mathbf{I})$, so

$$
\begin{aligned}
V(p_a, \boldsymbol{\mu}_a, \boldsymbol{\theta}_{\text{gen}}, \boldsymbol{\theta}_{\text{enc}}) = \quad & \mathbb{E}_{\boldsymbol{x}\sim p_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x}, E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}})) - c)^2] \\
& + (1-p_a)\mathbb{E}_{\boldsymbol{z}\sim N(\boldsymbol{z}|\mathbf{0},\mathbf{I})}[(D(G(\boldsymbol{z};\boldsymbol{\theta}_{\text{gen}}), \boldsymbol{z}) - c)^2] \\
& + p_a\mathbb{E}_{\boldsymbol{t}\sim N(\boldsymbol{t}|\mathbf{0},\mathbf{I})}[(D(G(\sigma_a\boldsymbol{t} + \boldsymbol{\mu}_a;\boldsymbol{\theta}_{\text{gen}}), \sigma_a\boldsymbol{t} + \boldsymbol{\mu}_a) - c)^2] \\
& + \mathbb{E}_{\boldsymbol{x}\sim p^a_{\text{real}}(\boldsymbol{x})}[(D(\boldsymbol{x}, E(\boldsymbol{x};\boldsymbol{\theta}_{\text{enc}})) - c)^2] \\
\approx \quad & \frac{1}{N_1}\sum_{i=1}^{N_1}(D(\tilde{\boldsymbol{x}}_i, E(\tilde{\boldsymbol{x}}_i;\boldsymbol{\theta}_{\text{enc}})) - c)^2 \\
& + (1-p_a)\frac{1}{N_2}\sum_{i=1}^{N_2}(D(G(\tilde{\boldsymbol{z}}_i;\boldsymbol{\theta}_{\text{gen}}), \tilde{\boldsymbol{z}}_i) - c)^2 \\
& + p_a\frac{1}{N_3}\sum_{i=1}^{N_3}(D(G(\sigma_a\tilde{\boldsymbol{t}}_i + \boldsymbol{\mu}_a;\boldsymbol{\theta}_{\text{gen}}), \sigma_a\tilde{\boldsymbol{t}}_i + \boldsymbol{\mu}_a) - c)^2 \\
& + \frac{1}{N_4}\sum_{i=1}^{N_4}(D(\tilde{\boldsymbol{x}}'''_i, E(\tilde{\boldsymbol{x}}'''_i;\boldsymbol{\theta}_{\text{enc}})) - c)^2
\end{aligned} \tag{50}
$$

Based on this estimation formula, the parameter gradients are estimated and then updated. Here, $\tilde{\boldsymbol{x}}_i$ are samples randomly drawn from the distribution $p_{\text{real}}(\boldsymbol{x})$, with a total of N_1 samples drawn. The random sampling method is to randomly draw samples $\tilde{\boldsymbol{x}}_i$ from the contaminated training dataset. $\tilde{\boldsymbol{z}}_i$ are samples randomly drawn from the distribution $N(\boldsymbol{z}|\mathbf{0}, \mathbf{I})$, with a total of N_2 samples drawn. The random sampling method is to use an algorithm to randomly draw samples $\tilde{\boldsymbol{z}}_i$ from the Gaussian distribution $N(\cdot|\mathbf{0}, \mathbf{I})$. $\tilde{\boldsymbol{t}}_i$ are samples randomly drawn from the distribution $N(\boldsymbol{t}|\mathbf{0}, \mathbf{I})$, with a total of N_3 samples drawn. The random sampling method is to use an algorithm to randomly draw samples $\tilde{\boldsymbol{t}}_i$ from the Gaussian distribution $N(\cdot|\mathbf{0}, \mathbf{I})$. $\tilde{\boldsymbol{x}}'''_i$ are samples randomly drawn from the distribution $p^a_{\text{real}}(\boldsymbol{x})$, with a total of N_4 samples drawn. The random sampling method is to randomly draw samples $\tilde{\boldsymbol{x}}'''_i$ from the auxiliary dataset.

D Processing of Parameter p_a

Since the parameter p_a is a probability value with a range in the closed interval $[0, 1]$, for better optimization, we can use the parameter s_a to represent p_a: $p_a = \frac{1}{s_a}$, where s_a has a range of $[1, +\infty)$, and then optimize the parameter s_a instead. Since p_a represents the model's prior anomaly probability and the anomaly probability usually should not be too large, we set a maximum anomaly probability $p_a^{\max}$, which is a hyperparameter, for example 0.1. Then, expressing p_a in terms of s_a can be changed to: $p_a = \frac{1}{\frac{1}{p_a^{\max}} + s_a}$, with s_a in the range $[0, +\infty)$, and during training we still optimize s_a. To ensure s_a is not less than 0, when optimizing s_a, after updating s_a using gradient descent or Adam, we check whether $s_a < 0$ and set s_a to 0 if so. Additionally, we set a hyperparameter p_{a0}, which is the initial value of p_a, and correspondingly, the initial value of s_a is $\frac{1}{p_{a0}} - \frac{1}{p_a^{\max}}$.

E Detailed Dataset Information

1. F-MNIST[5]: Fashion-MNIST is a classification dataset containing 70,000 grayscale images of size 28×28, covering 10 fashion classes (e.g., T-shirts, trousers, shoes, etc.). The training set includes 60,000 images, and the test set includes 10,000 images.
2. CIFAR-10[6]: CIFAR-10 is a classification dataset containing 60,000 color images of size 32×32, covering 10 classes such as airplanes, cars, birds, etc. The training set includes 50,000 images, and the test set includes 10,000 images.
3. OCT[7] [13]: Optical coherence tomography images selected from a retrospective cohort of adult patients, comprising 84,495 X-ray images, each belonging either to the normal class or to one of three abnormal disease classes.
4. ISIC[8] [17]: A public medical image dataset for skin lesions, where the nevus (NV) class serves as normal class and the remaining 6 classes are abnormal, used for anomaly detection research on skin lesion images.
5. Chest X-Ray[9] [30]: A dataset containing 5,856 chest X-ray images of pediatric patients, including normal and pneumonia cases. There are 1,349 normal images for training, and 234 normal images plus 390 abnormal images for testing.

The summary of dataset specifications is presented in Table 2.

Table 2: Overview of the datasets used.

Dataset	# of ano. classes	Training		Testing	
		# of norm.	cont. ratio	# of norm.	# of ano.
F-MNIST	9	6000	[0,0.2]	1000	9000
CIFAR-10	9	5000	[0,0.2]	1000	9000
OCT	3	26315	[0,0.2]	250	750
ISIC	6	6705	[0,0.2]	909	603
Chest X-ray	1	1349	[0,0.2]	234	390

F Processing of Datasets

The contaminated dataset is a mixture of normal samples from the original training set and a small number of anomalous samples randomly selected from the original training set. The auxiliary dataset consists of a small number of anomalous samples randomly selected from the original training set, and it is

[5] https://github.com/zalandoresearch/fashion-mnist
[6] https://www.cs.toronto.edu/ kriz/cifar.html
[7] https://data.mendeley.com/datasets/rscbjbr9sj/2
[8] https://challenge.isic-archive.com/data/#2018
[9] https://data.mendeley.com/datasets/rscbjbr9sj/2

aware that all samples in the auxiliary dataset are anomalous during training. The validation and test sets are obtained by splitting the original test set. The ELITE method additionally requires a labeled normal sample dataset, so we provide an additional labeled normal dataset of the same size as the labeled anomalous dataset. k represents the number of classes of labeled anomalous samples, and also the number of classes of labeled anomalous samples in the validation set. This is because the number of anomalous classes that can be seen during training should only be k, and the validation set should not contain any other classes of anomalous samples. Furthermore, all images are scaled to 32×32.

G Information of Model Training

Our model was trained using Adam algorithm [14], with learning rates of 0.0001, 0.0001, 0.000025, 0.001, and 0.001 for $\boldsymbol{\theta}_{\mathrm{enc}}$, $\boldsymbol{\theta}_{\mathrm{gen}}$, $\boldsymbol{\theta}_d$, s_a, and $\boldsymbol{\mu}_a$, respectively. $p_a^{\max}$ was set to 0.1, and p_{a0} was set to 0.05. The architectures of the neural networks of UCA-GAN are the same as ones of CIBiGAN [25].